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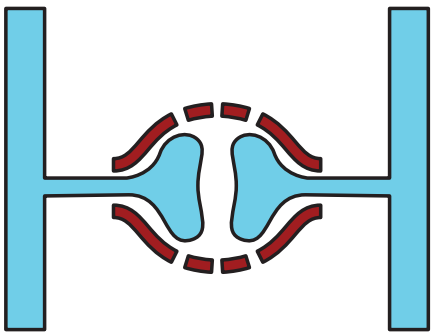
fungi, stink horns, rusts and smuts (Figure 1.23).



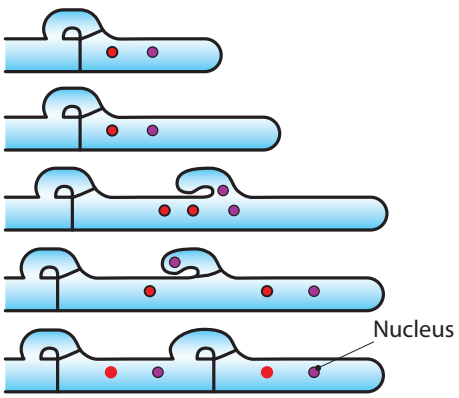
(a) *Agaricus*



(b) *Geaster*



(c) Dolipore septum



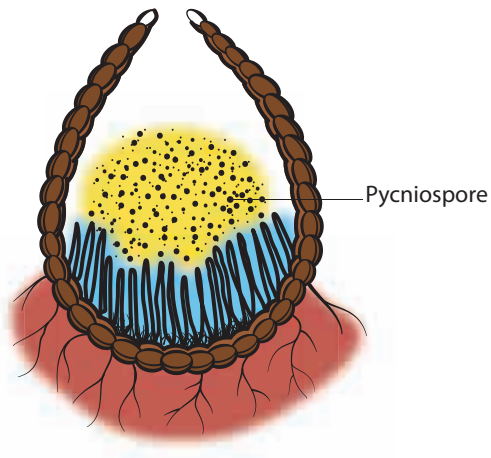
(d) Clamp connection

Figure 1.23: Structure and Reproduction in Basidiomycetes

- The members are terrestrial and lead a saprophytic and parasitic mode of life.
- The mycelium is well developed, septate with dolipore septum(bracket like). Three types of mycelium namely Primary (Monokaryotic), Secondary (Dikaryotic) and tertiary are found.
- Clamp connections are formed to maintain dikaryotic condition.
- Asexual reproduction is by means of conidia, oidia or budding.
- Sexual reproduction is present but sex organs are absent. Somatogamy or spermatisation results in plasmogamy. Karyogamy is delayed and dikaryotic phase is prolonged. Karyogamy takes place in basidium and it is immediately followed by meiotic division.
- The four nuclei thus formed are transformed into basidiospores which are borne on sterigmata outside the basidium (Exogenous). The basidium is club shaped with four basidiospores, thus this group of fungi is popularly called "Club fungi". The fruit body formed is called Basidiocarp**.**

Deuteromycetes or Fungi Imperfecti

The fungi belonging to this group lack sexual reproduction and are called imperfect fungi. A large number of species live as saprophytes in soil and many are plant and animal parasites. Asexual reproduction takes place by the production of conidia, chlamydospores, budding, oidia etc., Conidia are also produced in special structures called pycnidium, Acervulus, sporodochium and Synnema



(a) Pycnidium - *Phoma*